

Memory mapped I/O

Part of the memory address space dedicated to I/O devices.

$(E0000H - E0FFFH) - 4096 \text{ Mm}$

addresses assigned to I/O Ports

0000

U f f f

212

for words,

{ 0000H - E 0 F F H - 4096 Memory
 1 Port - 8 bit, 8 bit 1 location, is J.

1 Port - 8 bit, 8 bit 1 location, is 1

enough,

instructions that affect data in memory

(MOV, ADD, AND etc can be used To perform

I/O operations

I/O operations.

Is called mapped I/O - data transfer with
just accumulator.

mapped $\rightarrow$ AX, BX, CX any reg can be

just accumulator -
memory mapped $\rightarrow$ AX, BX, CX any reg can be
used.

Druckbuch

slow memory instruction

slow memory instructions
- memory instructions perform slower.

Isolated

use separate memory space

- limited inst.

IN, OUT, INS, OUTS

memory Address
space NOT offset

• mem mapped I/O

uses memory from the
-mm.

slower because memory
execute slower than the
special I/O

Minimum Mod interface



~~For~~
I/O devices can be present
LED's, switches, keyboard

I/O device & MPU

over the multiplexed
address data bus.

ADD - ADD, ADD - A'S

16 bit multiplexed.

ALE - store address
Value, (latch) so data
don't affect address.

$\overline{RD}$ - 0 tells the I/O that an input (read) is in progress.

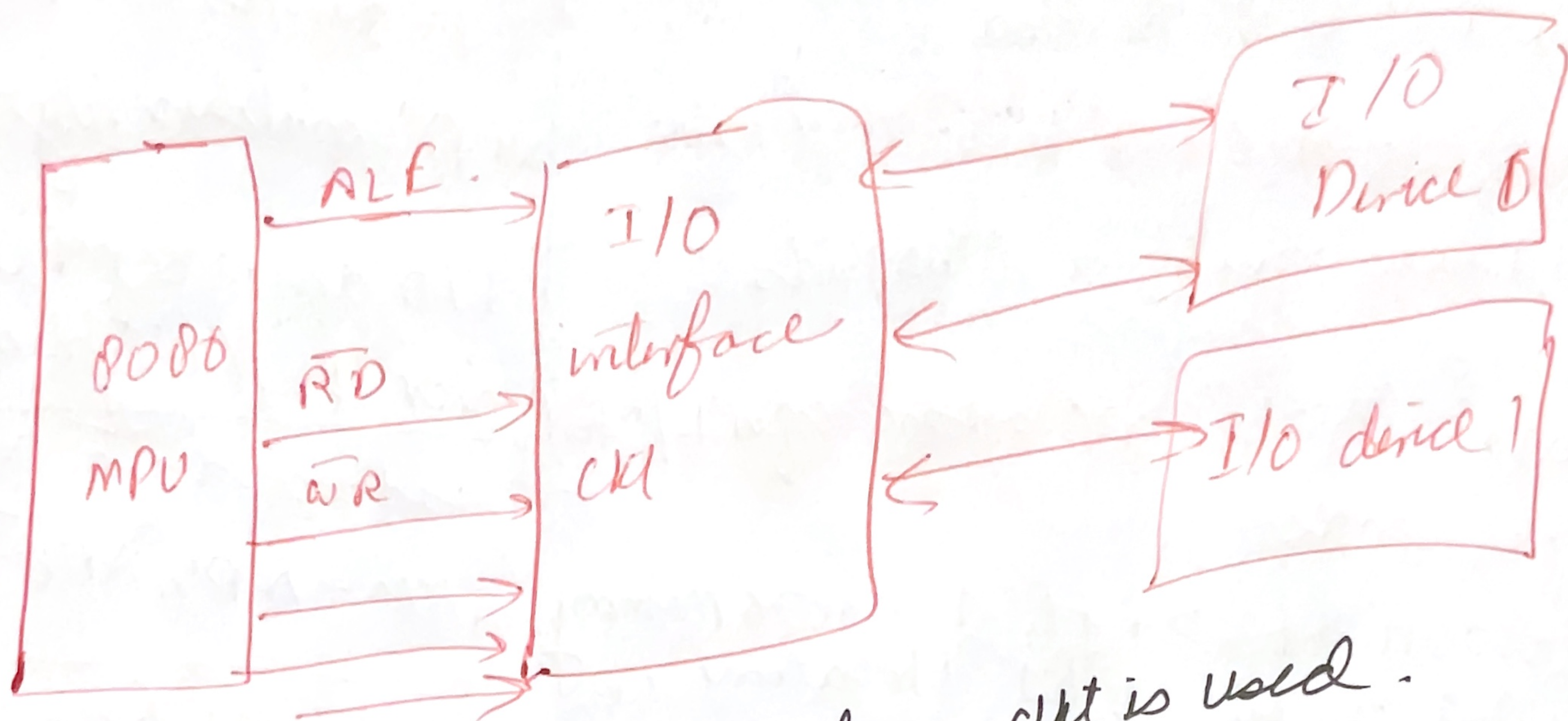
$\overline{wr}$ → tells the I/O interface that,

M/I/O $\rightarrow$ 0 - data

- transfer operation is for the IO subsystem

DT / $\bar{R}$ = Direction of data bus for input read / O/P write,

DEN - enables intstace



To interface I/O device, I/O interface chip is used.
Maximum-Mode interface.

8288 bus Controller produces the control signals for I/O subsystem.

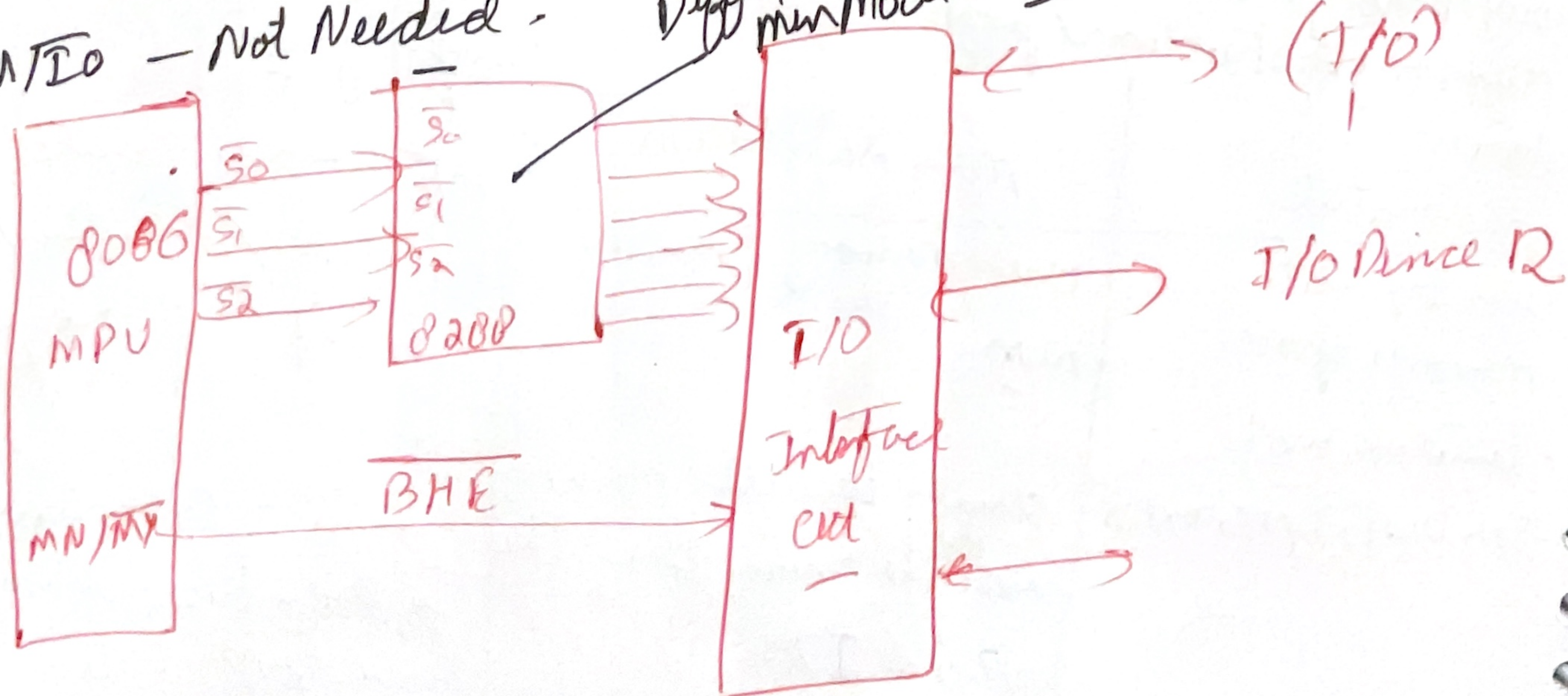
- signal changes

$\overline{IORC}$ replaces $\overline{RD}$
 $\overline{IOWC}$ & $\overline{AIOWC}$ replaces $\overline{WR}$.

$\overline{DEN}$ — complement of $\overline{DEN}$

$\overline{MTIO}$ — Not Needed.

Difference between Max Mode instruction



Type of IN/OUT instruction.

① Direct / Variable

↳ address is not given directly

To transfer a byte or a word of data.

No flags affected

IN AL, 0C8H

IN AX, 34H ; // 2 byte → from port 34H, 35H to AX

OUT 3BH, AL ; copy the contents of the AL to port 3BH

OUT 2CH, AX ; contents of AX to port 2CH, 2DH.

For a Variable port IN instruction the port address is loaded in DX Register before IN

0000 - ffff

Write a series of instructions that will O/P ffff H to an O/P port located at address B000H of the I/O address space

MOV AX, ffffH

~~MOV~~

OUT B000H, AX

or MOV AX, ffffH

MOV DX, B000H

OUT DX, AX

Data are to be read from

2 byte wide input port

at address, AAH, A9H

and then O/P a word

to a word wide O/P port

at address B000H

① Data Read ② O/P

IN AX, A9H

& wait address location

A9 & AA read 16 bit data -

into Accumulator

OUT B000H, AX

- ① we will put address.
- ② $ALE = \text{high}$ (latch address)
- ③ $M/\overline{IO} = 0$, To tell its a I/O operation.
- ④ $\overline{RD}$ - Read operation when 0, got data.

for write operation

$DT/\overline{R}$, high - write operation

$\overline{DEN} = \text{enable}$

Interface device

MP Need To interact with outside world, it needs interfacing ckt, establish a communication interface b/w MP & a world

- ① To match I/O device data & MP's data, (interfacing) device

Intel 8251: Universal

synchronous/asynchronous

Receiver Transmitter

8257: Programmable DMA Controller

Direct Mem Access:

When peripheral says it wants to communicate with memory, & MP (ask MP to give it access of that buses, we call it DMA-

I/O device asking for buses.

8259: Programmable Interrupt Controller

- interrupt to processor.
- single chip.
- 28 pin dual in line Package
- NMOS

8272

Intel 8275 Programmable CRT Controller

Q.1 enables MP To Read Data from

external devices.

(a) Printer ✓ ~~?~~ ~~XX~~

(d) Reader ✓

Q.2 while performing Read Operation, that much current should Not source / sink from data line.

Q.3 To avoid loading during Read op

(a) latch ~~✓~~

(d) tristate buffer

in Mem Mapped device are viewed as

(b) Memory locations

8086 does not support direct BCD Packed multiplication

Unpack → DAA → Pack

BCD → Binary Conversion

for 8086, the stack seg may have a memory block of

→ 2k - ? ss (memory = 64k)

Code stack Data Extra

A = 2000 something